ACME Integrated Delivery System

Hospital Nursing Intervention Pilot Program

Data Analysis Report

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Background

The ACME Integrated Delivery System (IDS) is a network of healthcare organizations under a parent holding company, comprising a network of physicians and hospitals. Leadership has identified an opportunity to improve patient outcomes by increasing nurse-to-patient ratios in hospitals with Intensive Care Units (ICU) and Surgical Intensive Care Units (SICU). Medical literature consistently demonstrates that higher nurse-to-patient ratios lead to better outcomes in intensive care settings.

However, this intervention will only be cost-effective at hospital sites with sufficient ICU and SICU bed volume, as measured by three bed metrics: licensed beds (total beds allowed by state license), census beds (average daily occupied beds), and staffed beds (beds for which clinical staffing exists). Leadership is interested in selecting one or two hospitals as pilot sites for the intervention, to be launched in the next fiscal quarter.

As a Data Analyst in the Business Insights Group, I was tasked with analyzing the ACME hospital network data to identify the best pilot site candidates. This report covers the database design, data quality validation, dimensional modeling, normalization assessment, and the analytical findings that support my final recommendation.

Dimensional Modeling

In dimensional modeling, variables are classified as either dimensions or facts. Dimensions are categorical or descriptive attributes used to filter, group, or label data. Facts are numeric, measurable quantities that can be aggregated to answer business questions. Understanding this distinction is essential for designing a star schema and writing correct SQL queries. The classification of every variable across all three tables is documented below.

bed_type Table (Dimension Table)

The bed_type table contains 20 rows describing the types of beds found across ACME's hospital network. All three variables are dimensions. None are facts, because nothing in this table represents a measurable, aggregable quantity.

Variable	Classification	Rationale
bed_id	Dimension (PK)	Unique numeric identifier for each bed type. Serves as the primary key and join key to the fact table.
bed_code	Dimension	Short alphabetic code categorizing the bed type (e.g., "IC" for ICU, "SI" for SICU).
bed_desc	Dimension	Human-readable description of the bed type (e.g., "ICU", "SICU", "Burn").

business Table (Dimension Table)

The business table contains 22,202 rows representing individual healthcare facilities. It includes both dimensions and facts. The three dimensions are:

Variable	Classification	Rationale
ims_org_id	Dimension (PK)	Unique identifier for each hospital. Serves as the primary key and join key to the fact table.
business_name	Dimension	The name of the hospital, a categorical label used for grouping and display.
bed_cluster_id	Dimension	A grouping code classifying facilities into size-based clusters. Despite being numeric, it is categorical.

The remaining three variables (ttl_license_beds, ttl_census_beds, ttl_staffed_beds) are facts. They are numeric totals that can be summed or averaged across hospitals.

bed_fact Table (Fact Table)

The bed_fact table contains 51,458 rows. Each row represents a specific bed type at a specific hospital. The three fact variables are:

Variable	Classification	Rationale
license_beds	Fact	Number of state-licensed beds of a given type at a given hospital.
census_beds	Fact	Average daily occupied beds for that bed type at that hospital.
staffed_beds	Fact	Number of beds with assigned clinical staff for that bed type.

The two remaining variables (ims_org_id and bed_id) are dimensions within the fact table. They are foreign keys linking each row to the business and bed_type dimension tables.

Normalization Assessment

Before building the star schema, I assessed the normalization level of each source table. Normalization is the process of organizing data to minimize redundancy and ensure that each piece of information is stored in exactly one place. Data that is properly normalized is cleaner, easier to maintain, and less prone to inconsistencies. The three standard levels of normalization (called normal forms) each build on the previous one.

First Normal Form (1NF)

First Normal Form requires that every column contains a single atomic value. There can be no repeating groups, no lists crammed into one cell, and no multi-valued fields. All three tables pass 1NF. Every cell in bed_type, business, and bed_fact contains exactly one value. For example, business_name holds a single hospital name per row, not a comma-separated list of names. bed_id holds a single integer, not a range or a set of IDs.

Second Normal Form (2NF)

Second Normal Form requires that every non-key column depends on the entire primary key, not just part of it. This only matters for tables with composite primary keys. The bed_type table has a single-column primary key (bed_id), so bed_code and bed_desc each depend fully on bed_id. The business table also has a single-column primary key (ims_org_id), so all columns depend fully on it. The bed_fact table has a composite key of (ims_org_id, bed_id), and each bed count (license_beds, census_beds, staffed_beds) depends on the combination of both the hospital and the bed type. You need both to determine the number. All three tables pass 2NF.

Third Normal Form (3NF)

Third Normal Form requires that no non-key column depends on another non-key column. Every non-key attribute must depend directly on the primary key and nothing else. In the bed_type table, bed_code and bed_desc both depend directly on bed_id and not on each other (knowing the code does not determine the description without the key). In the bed_fact table, the three bed counts depend directly on the composite key. In the business table, business_name and bed_cluster_id depend directly on ims_org_id. The total bed columns (ttl_license_beds, ttl_census_beds, ttl_staffed_beds) could theoretically be derived by summing values in bed_fact, which would make them transitive dependencies. However, they are stored as independent pre-aggregated values, and in practice this is an accepted convention in dimension tables. All three tables pass 3NF.

Why No Additional Normalization Was Needed

Because the source CSV files were already normalized to 3NF, no restructuring or decomposition of the tables was necessary before building the star schema. Each table stores one type of information with no redundancy: bed_type stores bed classifications, business stores hospital information, and bed_fact stores the actual bed count measures. Hospital names are not repeated in the fact table. Bed

descriptions are not repeated in the fact table. Instead, the fact table references both dimension tables through foreign keys (ims_org_id and bed_id), which is exactly how a properly normalized star schema should be structured. This meant I could proceed directly to defining keys, validating data quality, and running analysis queries.

Star Schema Design

Table Creation and Data Import

The three tables were created and populated using the MySQL Workbench Table Data Import Wizard. After connecting to the local MySQL server instance and creating a new database, I right-clicked the database name in the Navigator panel and selected "Table Data Import Wizard." The wizard reads the CSV headers and automatically creates a table with matching column names, detecting data types based on the values in each column (integers for bed counts, text for names and IDs).

Each CSV file was imported separately: bed_type.csv produced the bed_type table (20 rows), business.csv produced the business table (22,202 rows), and bed_fact.csv produced the bed_fact table (51,458 rows). The import order did not matter at this stage because foreign key constraints had not yet been defined.

One issue encountered during this process was that the Import Wizard assigned the TEXT data type to the ims_org_id column instead of VARCHAR. TEXT columns cannot be used as primary keys or foreign keys in MySQL, so ALTER TABLE MODIFY statements were used to convert these columns to VARCHAR(20) before adding any key constraints.

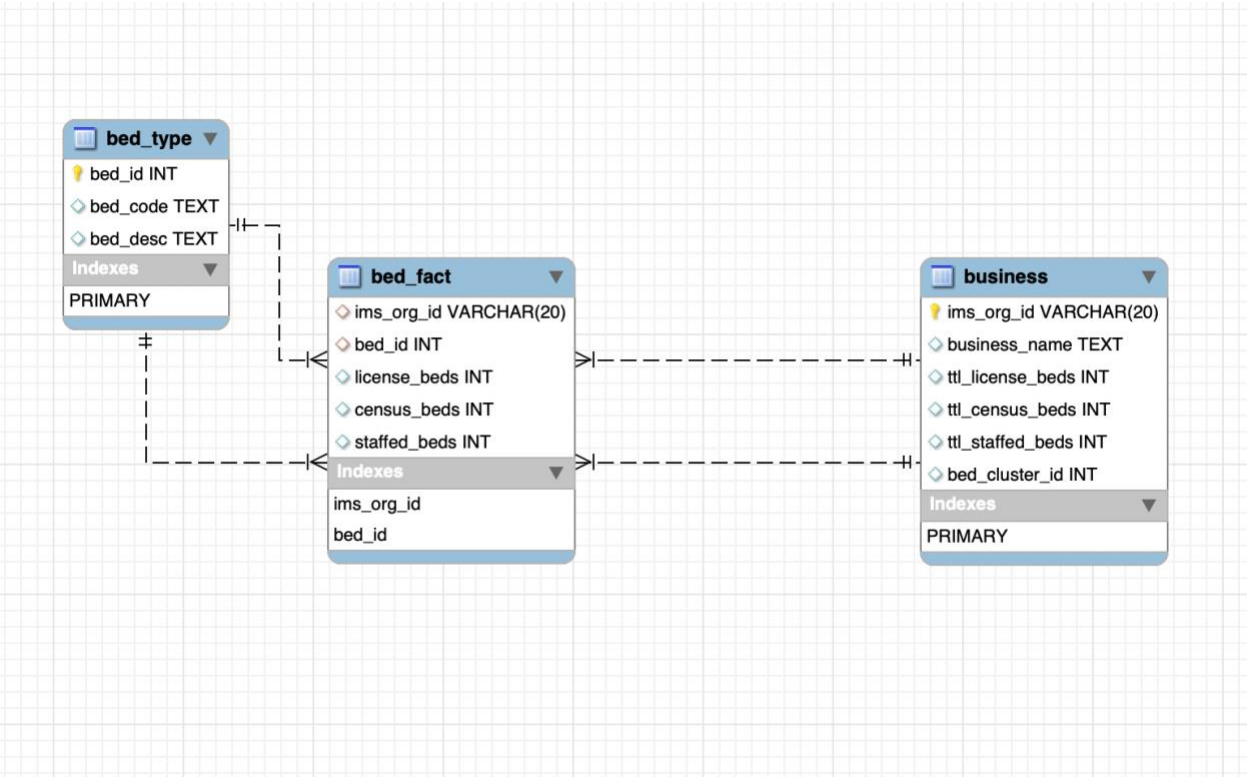
Schema Structure

The star schema places the bed_fact table at the center, storing granular bed counts at the intersection of each hospital and bed type. Two dimension tables radiate outward: business provides hospital names and identifiers, while bed_type provides bed type descriptions and codes. The foreign keys ims_org_id and bed_id in the fact table join to their respective dimension table primary keys. This design allows filtering by bed type (ICU, SICU) and aggregating by hospital name in a single JOIN query.

Primary keys were added to both dimension tables (bed_id in bed_type, ims_org_id in business) using ALTER TABLE ADD PRIMARY KEY statements. Foreign keys were then added to the fact table referencing those primary keys. This enforces referential integrity, meaning the database will reject any fact table row that references a hospital or bed type that does not exist in the corresponding dimension table. The foreign keys must be added after the primary keys exist, and the dimension tables must already contain data so that every foreign key value in bed_fact has a matching record in the parent table.

EER Diagram

The star schema was visualized using MySQL Workbench's Reverse Engineer feature. From the top menu, I selected Database, then Reverse Engineer, chose the active connection, selected the project database, and completed the wizard. MySQL Workbench generated an Entity Relationship (EER) diagram showing all three tables with their columns and data types. Because primary keys and foreign keys were already defined, the diagram automatically drew relationship lines between the foreign key columns in bed_fact and the primary key columns in business and bed_type. Text labels were added above each table to identify which is the fact table and which are dimension tables. The diagram was exported as a PNG image using File, Export, Export as PNG.



Data Quality Validation

Before running any analysis, I validated the data to ensure the results would be trustworthy. Data quality issues like NULL values, duplicate records, or orphan foreign keys can silently corrupt aggregated results. The following checks were performed using SQL queries.

Row Counts

The business table contains 22,202 rows, the bed_type table contains 20 rows, and the bed_fact table contains 51,458 rows. All counts match the expected data volume from the source CSV files.

NULL Values

A check for NULL values across all columns in all three tables returned zero NULLs. Every row has complete data, which means no records will be silently dropped during JOIN operations or aggregations.

Duplicate Records

A check for duplicate primary key values returned zero duplicates in both the business table (ims_org_id) and the bed_type table (bed_id). The bed_fact table also has zero fully duplicated rows. This confirms that no hospital or bed type is double-counted.

Orphan Records

An orphan record is a row in the fact table whose foreign key value has no matching primary key in the parent dimension table. A LEFT JOIN check confirmed zero orphan records in both directions: every ims_org_id in bed_fact exists in business, and every bed_id in bed_fact exists in bed_type. This means the foreign key relationships are complete and no fact rows will be lost during analysis joins.

All validation checks passed. The data is clean, complete, and ready for analysis.

Analysis: ICU or SICU Bed Volume

The first round of analysis identifies the top 10 hospitals by ICU (bed_id = 4) or SICU (bed_id = 15) bed volume. A hospital qualifies if it has at least one of either bed type. The SQL queries join bed_fact to the business dimension table, filter for ICU and SICU bed types, sum the bed counts per hospital, and return the top 10 in descending order. Three separate reports are provided, one for each bed metric.

Top 10 by Licensed Beds

hospital_name	total_license_beds
Phoenix Childrens Hospital	247
University of Maryland Medical Center	220
UC Health University Hospital	218
Wesley Medical Center, LLC	214
Vidant Medical Center	204
Rady Childrens Hospital and Health Center	200
Dallas County Hospital Association	195
Saint Lukes Episcopal Hospital Texas Medical C…	178
The Methodist Hospital	170
Emory University Hospital	169

Top 10 by Census Beds

hospital_name	total_census_beds
Shands Hospital at the University of Florida	167
Dallas County Hospital Association	145
Mercy Medical Center Saint Louis	142
Los Angeles County University of Southern Calif…	139
The Methodist Hospital	138
University of Minnesota Medical Center Fairview	129
University of Maryland Medical Center	127
Brigham and Womens Hospital	124
Vidant Medical Center	123
Ronald Reagan University of California Los Ang…	122

Top 10 by Staffed Beds

hospital_name	total_staffed_beds
Vidant Medical Center	203
Rady Childrens Hospital and Health Center	200
University of Maryland Medical Center	171
Emory University Hospital	169
Shands Hospital at the University of Florida	167
Mercy Medical Center Saint Louis	163
Wesley Medical Center, LLC	162
Phoenix Childrens Hospital	159
Grady Memorial Hospital	154
Los Angeles County University of Southern Calif…	151

Interpretation

Cross-List Consistency

Two hospitals appear on all three top-10 lists: University of Maryland Medical Center and Vidant Medical Center. This consistency across licensed, census, and staffed bed measures indicates that these facilities maintain high ICU/SICU capacity that is not only licensed but also actively occupied and staffed with clinical personnel.

Licensed Capacity vs. Actual Utilization

Phoenix Childrens Hospital leads the licensed bed ranking with 247 beds but does not appear on the census bed list, suggesting that its average daily occupancy is lower than its licensed capacity. A similar pattern is observed for Wesley Medical Center (214 licensed beds, absent from census top 10). For the pilot program, Leadership should prioritize hospitals where high licensed capacity translates into high actual patient volume.

High-Utilization Standout

Shands Hospital at the University of Florida leads the census ranking (167 beds) and matches that figure exactly on the staffed bed list (167 staffed beds), suggesting near-complete utilization of its staffed ICU/SICU capacity.

Drill-Down: Hospitals with Both ICU and SICU

The second round narrows the focus to hospitals operating both ICU and SICU beds. Out of 3,372 hospitals with at least one ICU or SICU bed, only 179 (approximately 5.3%) operate both bed types. The SQL queries add subquery filters ensuring each hospital has at least 1 ICU bed and at least 1 SICU bed before summing the combined totals.

Top 10 by Licensed Beds (Both ICU & SICU)

hospital_name	total_license_beds
University of Maryland Medical Center	220
UC Health University Hospital	218
Shands Hospital at the University of Florida	167
MCGHealth, Inc	155
Grady Memorial Hospital	154
Jackson Memorial Hospital	151
University of Minnesota Medical Center Fairview	144
University Hospital in Bexar County	144
Carolinas Medical Center	137
Yale New Haven Hospital	136

Top 10 by Census Beds (Both ICU & SICU)

hospital_name	total_census_beds
Shands Hospital at the University of Florida	167
University of Minnesota Medical Center Fairview	129
University of Maryland Medical Center	127
Jackson Memorial Hospital	117
UC Health University Hospital	110
Carolinas Medical Center	106
Cedars Sinai Health System	92
Duke University Health System	91
University Hospital in Bexar County	91
Grady Memorial Hospital	88

Top 10 by Staffed Beds (Both ICU & SICU)

hospital_name	total_staffed_beds
University of Maryland Medical Center	171
Shands Hospital at the University of Florida	167
Grady Memorial Hospital	154
UC Health University Hospital	151
University of Minnesota Medical Center Fairview	144
Carolinas Medical Center	137
Saint Josephs Hospital and Medical Center	134
Chattanooga Hamilton County Hospital Authority	134
Jackson Memorial Hospital	128
Sunrise Hospital and Medical Center, LLC	125

Utilization Rate Analysis

Raw bed counts tell part of the story, but they do not reveal how efficiently a hospital uses its capacity. A hospital with 220 licensed beds and 127 census beds has a very different operating profile than one with 167 licensed beds and 167 census beds. To capture this, I calculated two additional metrics for hospitals with both ICU and SICU beds:

Utilization rate is census beds divided by licensed beds, expressed as a percentage. It answers the question: of the beds this hospital is licensed to operate, what percentage are actually occupied on a typical day? A higher utilization rate means more patients are being served relative to capacity.

Staffing rate is staffed beds divided by licensed beds, expressed as a percentage. It answers the question: of the beds this hospital is licensed to operate, what percentage currently have clinical staff assigned? A gap between staffing rate and 100% indicates untapped capacity that could be activated with additional hiring.

To avoid misleading percentages from very small hospitals (a 2-bed facility at 100% is not meaningful for a pilot program), this analysis is filtered to hospitals with at least 50 licensed ICU/SICU beds.

Top 10 by Utilization Rate (Minimum 50 Licensed Beds)

Hospital Name	Licensed	Census	Staffed	Utilization	Staffing Rate
Shands Hospital at the Univ. of Florida	167	167	167	100.0%	100.0%
Sharp Metropolitan Medical Campus	64	62	64	96.9%	100.0%
UC Davis Medical Center	50	48	48	96.0%	96.0%
Cedars Sinai Health System	96	92	94	95.8%	97.9%
Arrowhead Regional Medical Center	66	62	66	93.9%	100.0%
Buffalo General Hospital	57	53	57	93.0%	100.0%

Saint Barnabas Medical Center	94	86	91	91.5%	96.8%
North Shore University Hospital	77	70	77	90.9%	100.0%
Harborview Medical Center	91	82	91	90.1%	100.0%
Beaumont Hospital Royal Oak	80	72	75	90.0%	93.8%

Recommended Hospitals: Utilization Comparison

For context, here is how the two recommended pilot hospitals compare on utilization:

Hospital Name	Licensed	Census	Staffed	Utilization	Staffing Rate
Univ. of Maryland Medical Center	220	127	171	57.7%	77.7%
Shands Hospital at the Univ. of Florida	167	167	167	100.0%	100.0%

This comparison highlights why both hospitals are valuable as a pair. UMMC has the larger overall capacity (220 licensed beds) but operates at 57.7% utilization, meaning there is significant room to grow. Its 77.7% staffing rate also reveals a gap between licensed and staffed capacity that the nursing intervention could directly address. Shands, on the other hand, runs at 100% across the board. Every licensed bed is staffed and every staffed bed is occupied. The intervention at Shands would test whether adding nurses above the current staffing level improves outcomes in an already fully utilized unit.

Final Recommendation

Based on the combined findings from the bed volume analysis, the drill-down investigation, and the utilization rate analysis, I recommend the following two hospitals as pilot sites for the ACME Nursing Intervention Program:

1. University of Maryland Medical Center

UMMC is the only hospital to appear on all six top-10 bed volume lists across both rounds of analysis. In the drill-down (both ICU and SICU required), it ranked 1st in licensed beds (220) and staffed beds (171), and 3rd in census beds (127). Its 77.7% staffing rate reveals a gap between licensed capacity and current staffing, which means the nursing intervention has clear room to make an impact. With the largest dual ICU/SICU operation in the dataset, the pilot would reach a high volume of critical care patients and generate robust data for evaluating the intervention.

2. Shands Hospital at the University of Florida

Shands ranked 1st in census beds (167), 2nd in staffed beds (167), and 3rd in licensed beds (167) in the drill-down analysis. All three figures are identical, producing a 100% utilization rate and a 100% staffing rate. This is the highest utilization of any hospital with 50 or more licensed ICU/SICU beds in the entire dataset. The intervention at Shands would impact the maximum number of patients per bed, and the consistently full units provide an ideal controlled environment for measuring clinical outcomes.

Why these two together: UMMC represents high capacity with room to grow (large bed count, moderate utilization, staffing gap to fill). Shands represents maximum utilization with no slack (every bed full, every bed staffed). Running the pilot at both sites tests the intervention under two fundamentally different operating conditions. If outcomes improve at both a high-capacity site and a high-utilization site, Leadership will have strong evidence to support a network-wide rollout.